Module 3 Homework 3 Challenge

Below is a scored review of your assessment. All questions are shown.

Earned 59,994 of 99,990 points. 60%

Main Question Set

Solve six of the first ten questions to achieve a passing score of 60%. Don't miss the learning opportunity in the rest of the problems, though -- spend the full time and attempt every one of them!

Correct Answer Partially Correct Incorrect Answer

1

Tickets to a movie cost $\$7$ for adults and $\$5$ for kids. Suppose 50 total people went to a showing of the movie and $\$320$ total were spent. Let x equal the number of adults and y equal the number of kids. Which of the following sets of equations matches the information given?

Your Answer

$$7x + 5y = 320$$

$$x + y = 50$$

Correct Answer

$$7x + 5y = 320$$

$$x + y = 50$$

Explanation

Choice 1: $\$12 \times (x + y) = \$320$ doesn't make sense because it says that all the people, adults and kids combined, paid $\$12$ each for their tickets, whereas $\$12$ is really the cost of two tickets, one adult and one child.

Choice 2: The first equation, $\$7x + \$5y = \$320$, has its first term meaning the cost of all adult tickets, and the second term the cost of all kid tickets, which does add up to the total cost of all tickets, $\$320$. The second equation, $x + y = 50$, says that the number of all people (adults plus kids) is equal to 50, which is true.

Choice 3: The first equation, $\$7y + \$5x = \$320$, has the x and y swapped, so it is incorrect. The second equation, $xy = 50$, plainly differs from the correct one as well, $x + y = 50$.

Choice 4: The units of the left side in $\frac{\$320}{\$7} + \frac{\$320}{\$5} = 50$ come out to $\frac{\text{total dollars}}{\text{dollarsper kid}} = \text{number of kids}$. However, this assumes that all the money was spent by the kids, and that same amount was also spent by the adults once again, which isn't correct, because they share the $\$320$ expenditure.

Choice 5: The equation $\frac{\$320}{x} + \frac{\$320}{y} = \$12$ tells us that all the $\$320$ was spent by the kids, and then the total $\$320$ was also spent one more time by the adults. In reality, the $\$320$ should be split over the x and the y.

2 If $3x + 7y = 420$ and $x = \frac{7}{3}y$, then what is $x + y$?

Your Answer	100
Correct Answer	100

Explanation

Plug and chug! If $x = \frac{7}{3}y$, then $3x = 7y$, so plugging this into the first equation gives us $6x = 420$ and so $x = 70$. Then we have $y = \frac{3}{7}x = \frac{3}{7} \times 70 = 30$ and so $x + y = 70 + 30 = 100$.

3

When the decimal point of a positive number x is moved 2 places to the right, the new number is 9 times the **reciprocal** of the original number. Find the expression that accurately matches this.

Your Answer	$100x = \frac{9}{x}$
Correct Answer	$100x = \frac{9}{x}$

Explanation

Moving the decimal point 2 places to the right makes the number $10^2 = 100$ times larger. If we call the number x, we now have the equation $100x = 9/x$.

4 In the quadratic equation $x^2 + ax + b = 0$, where a and b are real numbers, which of the following correctly describes the relationship between a and b and the roots of $x^2 + ax + b = 0$?

Your Answer

$$a = (-1) \times \left(\text{sum of the roots}\right)$$
$$b = \text{product of the roots}$$

Correct Answer

$$a = (-1) \times \left(\text{sum of the roots}\right)$$
$$b = \text{product of the roots}$$

Explanation

Let's use what we know about the roots of quadratic functions to help us write the equations out. We know that the roots are the values of x that make the quadratic $x^2 + ax + b = 0$.

There can't be more than two roots, either. We know that we can then factor the quadratic as

$$x^2 + ax + b = (x - r_1)(x - r_2)$$

where r_1 and r_2 are the two roots of x. You can check that it works by seeing that the left factor, $x - r_1$, becomes 0 when $x = r_1$, and the right factor, $x - r_2$, becomes 0 when $x = r_2$. Since the factors are multiplied together, if either of them equals 0, the whole quadratic equals 0.

Expanding the factored equation out, we can see the values of a and b :

$$\begin{aligned} x^2 + ax + b &= (x - r_1)(x - r_2) \\ &= x^2 - r_1x - r_2x + (-r_1)(-r_2) \\ &= x^2 + \Big(-1 \times (r_1 + r_2) \Big) + r_1r_2 \end{aligned}$$

Then, a equals -1 times the sum of the roots, and b equals the product of the roots, which gives us choice 1.

5 Robert has fish, chickens, and dogs. There are 28 eyes total and 16 legs total. Also, there are twice as many fish as chickens. How many dogs are there?

✓	**Your Answer**	2
	Correct Answer	2

Explanation

Let us call the number of fish f, the number of chickens c, and the number of dogs d.

$$f = 2 \times c$$

Each animal has 2 eyes, so since there are 28 total eyes, there must be $\frac{28}{2} = 14$ animals. Then,

$$f + c + d = 14$$

But

$$f = 2 \times c$$

So

$$3 \times c + d = 14.$$

The number of legs is $4 \times d + 2 \times c$. Then,

$$\begin{aligned} 4 \times d + 2 \times c &= 16 \\ 2 \times d + c &= 8 \\ 6 \times d + 3 \times c &= 24 \end{aligned}$$

Then if $3 \times c + d = 14$, and $6 \times d + 3 \times c = 24$,

$$5 \times d + d + 3 \times c = 14 + 10$$

$$5 \times d = 10$$

So d, the number of dogs, is 2.

6

When the decimal point of a positive number x is moved 2 places to the right, the new number is 9 times the **reciprocal** of the original number. Find the value of x. Your answer should be a fraction in simplest form without spaces, e.g. 1/2

Your Answer	3/10
Correct Answer	3/10

Explanation

Using the equation from the previous question, $100x = \frac{9}{x}$, we can multiply x to both sides to get $100x^2 = 9$, which means that $x^2 = \frac{9}{100}$. Square-rooting both sides, we get that $x = \pm\frac{3}{10}$, and knowing that x has to be positive, $x = 3/10$.

7 Ann realizes that the price of a skirt is 80% of her money and the price of a blouse is 60% of her money. After her grandma gives her $\$16$, Ann will be able to buy both. How much money in dollars does Ann have in the beginning (before her grandma gives her money)? Your answer should be the number only, without the $\$$ sign.

✕	**Your Answer**	[no answer given]
	Correct Answer	40

Explanation

Let x be the amount of Ann's money (not including her grandma's money). Then the price of a skirt is 80% of x, which equals $0.8x$. The same is true for a blouse: its price equals $0.6x$. After Ann's grandma gives her $\$16$, Ann will have $x + 16$. On the other hand, she will be able to buy the skirt and blouse together, so we can write these equations:

$$\begin{aligned} x + 16 &= 0.8x + 0.6x \\ x + 16 &= 1.4x \\ 16 &= 0.4x \\ 0.4x &= 16 \\ x &= 40 \end{aligned}$$

8 Find which of the following set of equations has a unique solution; i.e. each variable has a solvable value:

Your Answer	[no answer given]
Correct Answer	$3x + 4y + z = 20$
	$y - z + 2x = 3$
	$\frac{1}{2}z + \frac{3}{4}y = 10 - x$

Explanation

Hopefully you didn't try to solve for x, y, and z! You can eyeball this one. Use the fact that we can solve a set of equations if the number of variables is the same as the number of equations. If we have more variables than equations, we can still determine a relationship between the variables, but we could choose any value for one of them, say x for example, and then get many different answers. If we have more equations than variables, this could also be solvable; however, there might be no solution, for example if we have $x = 0$ and $x = 3$.

Notice that every set of choices has three variables, x, y, and z. However, only the fourth choice has three equations; every other choice has fewer than three.

9 The sum of two numbers is 76, and the sum of 25% of the first number and 15% of the second number is 14. Find the difference between these two numbers.

✕	**Your Answer**	[no answer given]
	Correct Answer	24

Explanation

Let's call the two numbers a and b. We know that $a+b=76$, and 25% of a plus 15% of b equals 14, from which it follows that $0.25a+0.15b=14$. So we have obtained a system of two equations and two unknown variables:

$$\begin{aligned} a+b &= 76 \\ 0.25a+0.15b &= 14 \end{aligned}$$

Let's solve it!

We know how to solve equations with one unknown variable, so let's transform our system of two equations to just one equation with one variable, which we know how to solve.

We know that $0.25=\frac{1}{4}$, so let's multiply by 4 so that the coefficient of a becomes 1 :

$$\begin{aligned} a+b &= 76 \\ a+0.60b &= 56. \end{aligned}$$

Now we can cancel variable a by subtracting the lower equation from the upper equation:

$$\begin{aligned} (a-a)+(b-0.60b) &= 76-56 \\ a+0.60b &= 56. \end{aligned}$$

Which is the same as:

$$\begin{aligned} 0.40b &= 20 \\ a+0.60b &= 56. \end{aligned}$$

From the first equation, we can find $b=\frac{20}{0.4}=50$. When we find one of the two variables, we can put the number we just found into the same equation to find the remaining variable. The most easy way to do this is to put $b=50$ into equation $a+b=76$:

$$a+50=76 \implies a=76-50=26$$

Now we found that $a=26$ and $b=50$, so the difference between these two numbers is $(b-a)=50-26=24$.

10 If you plug the *root* of the quadratic equation $x^2 + ax + b$ into itself, i.e. substituting the root for the value of x, the value of the expression becomes

Your Answer	[no answer given]
Correct Answer	0

Explanation

A root of a quadratic equation is the value of x which makes the expression $x^2 + ax + b = 0$.

Earned 0 of 7 points. 0%

Bonus Challenge Questions

These optional questions will not count towards your final score. However, we strongly suggest trying them for the benefit of the challenge!

11 The sum of the squares of two positive numbers is 800. Their product is 112. What is the sum of these two positive numbers?

Your Answer	[no answer given]
Correct Answer	32

Explanation

If we call the integers x and y, then $x^2 + y^2 = 800$ and $xy = 112$.
Therefore,

$$x^2 + 2xy + y^2 = 800 + 2(112) = 800 + 224 = 1024.$$

Square-rooting both sides of the equation, we get

$$x + y = \sqrt{1024} = \sqrt{2^{10}} = 2^5 = 32.$$

12 James and John are swimming. James swims $2x - 4$ miles in $5x - 6$ hours. John swims $10 - 2x$ miles in $x + 3$ hours. If they are swimming at the same speed, then find the value of x.

Your Answer	[no answer given]
Correct Answer	4

Explanation

Let's translate the information from the problem into math:

$$\begin{aligned}
\text{James' speed} &= \text{John's speed} \\
\frac{\text{James' distance}}{\text{James' time}} &= \frac{\text{John's distance}}{\text{John's time}} \\
\frac{2x-4}{5x-6} &= \frac{-2x+10}{x+3} \\
(2x-4)(x+3) &= (-2x+10)(5x-6) \\
2x^2 + 6x - 4x - 12 &= -10x^2 + 12x + 50x - 60 \\
12x^2 - 60x + 48 &= 0 \\
x^2 - 5x + 4 &= 0
\end{aligned}$$

We can factor this equation:

$$x^2 - 5x + 4 = (x-1)(x-4)$$

So there are two possible values for x, 1 and 4. The value of $x = 1$ wouldn't work because then the number of miles James swims would be $2 \times 1 - 4 = -4$, which is negative. Then, our answer is $x = 4$.

13 If p and q are the two solutions to $x^2 + 13x - 39$, find $(p-q)^2$.

Your Answer	[no answer given]
Correct Answer	325

Explanation

Using the Quadratic Formula, the difference in the two roots is equal to

$$\begin{aligned}\frac{2\sqrt{13^2 + 4 * 39}}{2} &= \sqrt{169 + 156} \\ &= \sqrt{325}.\end{aligned}$$

Therefore, the square of this quantity is 325.

14 A picture with dimensions 3 inches by 4 inches is placed in a picture frame of uniform width around all sides, enlarging its area to 42 square inches. What is the width of the picture frame? Your answer should be a decimal rounded to the nearest tenth, without units.

Your Answer	[no answer given]
Correct Answer	1.5

Explanation

Don't forget that the overall width increases by twice the width of the frame! The same is true for the length. Since the picture frame wraps around the picture on all 4 sides, we can call the picture frame's width p, and the new larger dimensions of the picture plus the picture frame are $(3+2p)\times(4+2p)$, so we get a nice quadratic equation:

$$\begin{aligned}(3+2p)\times(4+2p)&=42.\\4p^2+6p+8p+12&=42\\4p^2+14p+12&=42\\2p^2+7p+6&=21\\2p^2+7p-15&=0\\(2p-3)(p+5)&=0\end{aligned}$$

We can only get zero as the product of two quantities if either of the quantities (or both) is equal to zero, which means that either $2p-3=0$ or $p+5=0$. If $p+5=0$, then p will be negative, which wouldn't make sense, because p is the width of the picture frame. Then $2p-3=0$, which means that $2p=3$, and $p=\frac{3}{2}=1.5$.

15 Suppose $f(x) = x^2 + 17x - 39$ and $g(x) = 4x + 1$. What is the sum of all possible values of x such that $g(f(x)) = 0$?

✕	**Your Answer**	[no answer given]
	Correct Answer	-17

Explanation

If $g(f(x)) = 0$, then $4(x^2 + 17x - 39) + 1 = 0$, therefore $4x^2 + 68x - 155 = 0$. The sum of all roots of this polynomial is then just $-68/4 = -17$.

16 What is the sum of all integers p such that $x^2 + px + 20$ can be factored into $(x - a)(x - b)$ where a and b are integers?

✕	**Your Answer**	[no answer given]
	Correct Answer	0

Explanation

The sum of the roots, a and b, equals $-p$, and the product of a and b equals 20. Since a and b are integers, they both have to be factors of 20, whether negative or positive. $20 = 2^2 \cdot 5$ has 6 factors, meaning 6 unordered factor pairs that multiply to 20 including negative numbers. These pairs are:

$$(20, 1), (-20, -1), (10, 2), (-10, -2), (5, 4), (-5, -4).$$

It actually doesn't matter exactly what they are, because p is the negative sum of the two numbers inside the parentheses. And when we add up the sums of the pairs of numbers in every of the 6 factor pairs, the negative factor pairs' sums cancel out with those of the positive factor pairs. Therefore the total sum of all possible values of p is just 0.

17 Suppose that $x - c$ is a factor of $x^2 + 2cx - 3$. Find the larger of the two values of c.

✕	**Your Answer**	[no answer given]
	Correct Answer	1

Explanation

Notice that what we have isn't an equation, but just an expression. You might wonder, what is there to solve with just an expression? Actually, the roots of an expression will make that expression equal to 0, so our equation is implicit in this problem. Since we have a variable for one of the roots of the expression, it makes sense to make a variable for the other root, d. Then

$$\begin{aligned} x^2 + 2cx - 3 &= (x - c)(x - d) \\ &= x^2 + x(-c + d) - cd \end{aligned}$$

Looking at the coefficients in our original expressions, we can see that

$$cd = 3$$

and

$$2c = -c + d$$

Then from the second equation, we get

$$d = 3c,$$

and, plugging that into the first equation, we get

$$\begin{aligned} c \times 3c &= 3 \\ c^2 &= 1 \end{aligned}$$

Then the two values of c are -1 and 1. The question asks for the larger value, so the answer is 1.

Earned 0 of 3 points. 0%

Super Bonus Questions

These questions are an extreme challenge! Some of the most interesting, creative and clever questions are the hardest ones. So try every single one. Even if you can't figure it out, you'll still be deepening your understanding of the important math concepts that underlie this course.

18 The number of owls varies inversely with the number of mice, and the number that you get from multiplying the number of owls with the number of mice is always the same. When there are $7x - 5$ owls, there are $4x + 8$ mice. When there are $6x - 6$ owls, there are $11x + 2$ mice. What is the product of the number of owls and the number of mice?

Your Answer	[no answer given]
Correct Answer	144

Explanation

Explanation: Since the product of the owls and the mice is always the same, we can write

$$\begin{aligned}(7x-5)(4x+8) &= (6x-6)(11x+2)\\ 28x^2+36x+40 &= 66x^2-54x-12\\ 14x^2+18x-20 &= 33x^2-27x-6\\ 19x^2-45x+14 &= 0\end{aligned}$$

Here we have a quadratic equation of the form ax^2+bx+c, so the Quadratic Formula will make it easy to find the value of the roots, x.

$$\begin{aligned}x &= \frac{-b \pm \sqrt{b^2-4ac}}{2a}\\ x &= \frac{45 \pm \sqrt{(-45)^2 - 4\times 19\times 14}}{2\times 19}\\ x &= \frac{45 \pm \sqrt{961}}{38}\\ x &= \frac{45 \pm 31}{38}\\ x &= 2, \frac{7}{19}.\end{aligned}$$

Since there are never fractional numbers of owls and mice, $x = 2$. Therefore, the product of the owls and the mice is

$$\begin{aligned} \frac{7x-5}{4x+8} &= 9 \times 16 \\ x &= 144. \end{aligned}$$

19 Solve for x in the equation

$$\left(\frac{5-x}{4-x}\right)-\left(\frac{6-x}{5-x}\right)=\left(\frac{7-x}{6-x}\right)-\left(\frac{8-x}{7-x}\right)$$

Your answer should be a decimal rounded to the nearest tenth.

✕	**Your Answer**	[no answer given]
	Correct Answer	5.5

Explanation

We can notice that the numbers in the numerator are counting up, as are the numbers in the denominator. Let's parametrize the expressions so that the numbers look smaller, by creating a variable called a. If we let $4-x=a-x$, then this will make our fractions look a little bit better, and it's nice, because this means we can solve for a general form of this expression.

The equation is

$$\begin{aligned}\left(\frac{5-x}{4-x}\right)-\left(\frac{6-x}{5-x}\right)&=\left(\frac{7-x}{6-x}\right)-\left(\frac{8-x}{7-x}\right)\\ \left(\frac{a+1-x}{a-x}\right)-\left(\frac{a+2-x}{a+1-x}\right)&=\left(\frac{a+3-x}{a+2-x}\right)-\left(\frac{a+4-x}{a+3-x}\right)\\ \left(\frac{1}{(a-x)(a+1-x)}\right)&=\left(\frac{1}{(a+2-x)(a+3-x)}\right)\end{aligned}$$

This means that

$$\begin{aligned}(a-x)(a+1-x)&=(a+2-x)(a+3-x)\\ a^2+a-ax-ax-x+x^2&=a^2+3a-ax+2a+6-2x-ax-3x+x^2\\ a^2-2ax+a-x+x^2&=a^2-2ax+5a+6-5x+x^2\end{aligned}$$

Canceling the a^2, x^2, and $-2ax$ from both sides, we get

$$\begin{aligned}a-x&=5a+6-5x\\ 0&=4a-4x+6\\ 4x&=4a+6\\ x&=a+\frac{3}{2}\end{aligned}$$

We actually solved for the general case where a can be anything, but in our case, since the first fraction has $5-x$ in the numerator, we actually want $a=4$. Plugging in this, we get

$$x = 4 + \frac{3}{2}$$
$$x = 5.5$$

20 Let $f_1 = \frac{x+1}{x-1}$. Let f_2 be formed by substituting $\frac{x+1}{x-1}$ for x in f_1, f_3 be formed by substituting $\frac{x+1}{x-1}$ for x in f_2, and so on and so forth. Find f_x for $x = 2019$.
Your answer should be a fraction in simplest form without spaces, e.g. 1/2

Your Answer [no answer given]

Correct Answer 1010/1009

Explanation

Let's start with finding f_2. For this, we just need to substitute $\frac{x+1}{x-1}$ in for x :

$$\begin{aligned} f_2 &= \frac{\frac{x+1}{x-1} + 1}{\frac{x+1}{x-1} - 1} \\ &= \frac{x+1+x-1}{x+1-x+1} \\ &= \frac{2x}{2} \\ &= x \end{aligned}$$

Wow! This means that we get x back again after running the function once, so running the function again will give us

$$f_3 = \frac{x+1}{x-1}$$

The value of the function f_n then flip-flops between x and $\frac{x+1}{x-1}$, with all the odd-numbered n corresponding to f equaling $\frac{x+1}{x-1}$, and all the even-numbered n corresponding to f equaling x. All we have to do is look at the *parity* of n (whether it is even or odd). Since 2019 is odd, then

$$\begin{aligned} f_{2019} &= \frac{x+1}{x-1} \\ &= \frac{2019+1}{2019-1} \\ &= \frac{2020}{2018} \\ &= \frac{1010}{1009} \end{aligned}$$

Summary

Software by

brillium

Version 11.2

Privacy Policy. Assessment content is copyright 2026, Expii, Inc..